

ECE371 Neural Networks and Deep Learning

Assignment 1

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Abstract: In this assignment, we utilize the pre-trained ResNet152 model as the input model for the training process. The dataset is divided into an 80% training set and a 20% test set. A comprehensive configuration file is set up to provide suitable parameters and training strategies for the model. Task one achieves an accuracy rate of over 90%, while task two, under the condition of a single main.py, also reaches an accuracy rate of over 80%.

Keywords: ResNet152 Index WarmUp CosineAnnealing

1 Introduction

This project presents the development and evaluation of a deep learning model for flower image classification using a pre-trained ResNet152 architecture. The dataset was split into training and validation sets, with extensive data augmentation techniques applied to enhance model robustness. A combined learning rate scheduling strategy, featuring exponential warmup followed by cosine annealing, was employed to optimize training efficiency. Through rigorous training and validation, the model achieved significant accuracy improvements, demonstrating the effectiveness of the chosen training strategies. The insights gained from this project highlight the importance of data augmentation and adaptive learning rate scheduling in improving the performance of deep learning models for image classification tasks.

2 Related Work

Image classification is a fundamental task in the field of computer vision and is often used as a benchmark to measure progress in computer vision technologies. In recent years, significant advancements have been made in this area, driven by the development of deep learning techniques. Some of the latest research and trends from 2023 include: Vision Transformers (ViT): Vision Transformers have demonstrated powerful performance in image classification tasks, capable of capturing complex patterns and long-range dependencies in images. For example, the Swin Transformer proposed by Liu et al. in 2021, which uses hierarchical vision transformers with shifted windows, has further enhanced performance. Soft Augmentation: In 2023, Liu et al. introduced Soft Augmentation, a technique that softens learning targets to enhance the effectiveness of data augmentation. This approach allows for more aggressive data augmentation strategies, thereby improving model robustness and calibration. Model Integration and Optimization: In the 2023 ImageNet benchmark, Model Soups (BASIC-L) emerged as the state-of-the-art model, demonstrating the potential of model integration and optimization strategies to boost image classification performance. Self-Supervised Learning: The application of self-supervised learning in image classification has also garnered increasing attention. For instance, a review by Shurrab and Duwairi in 2022 explored self-supervised learning methods in medical image analysis. This approach leverages unlabeled data to learn image representations, thereby reducing the reliance on large amounts of labeled data. Multi-Scale Feature Learning: In 2023, Olimov et al. proposed an image classification model based on continuous multi-

scale feature learning, which improves classification performance by learning features at different scales. These studies reflect the ongoing progress in the field of image classification, from traditional Convolutional Neural Networks (CNNs) to modern Vision Transformers and new strategies in data augmentation and self-supervised learning. The combination of these technologies provides new directions for future research, particularly in enhancing model generalization and efficiency.

3 Method

Data Preprocessing and Augmentation:

The code employs a series of data augmentation techniques to enhance the robustness and generalization of the model. This includes random cropping, flipping, rotation, color jittering, Gaussian blur, and random erasing. These techniques help in creating a more diverse training dataset, which is crucial for improving the model’s ability to handle real-world variations in images. Detailed augmentation involves random cropping to 224×224 pixels, random flipping both horizontally and vertically, rotation within a specified range, adjustments in brightness, contrast, saturation, and hue, application of Gaussian blur, and random erasing of patches within the image. These are applied using `torchvision.transforms` for a flexible augmentation pipeline, followed by normalization adapted to the mean and standard deviation of the data set[1].

Model Selection and Modification:

The project utilizes a pre-trained ResNet152 model as the base architecture. The last fully connected layer is modified to accommodate the specific number of classes in the flower dataset. This involves adding additional layers including a linear layer, ReLU activation, dropout for regularization, and another linear layer to output the class probabilities. This modification allows the model to adapt to the specific classification task at hand[2].

Optimizer and Learning Rate Scheduling:

The model is trained using the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.1, momentum of 0.9, and weight decay of 1×10^{-4} . The learning rate is managed using a combination of an exponential warmup phase and a cosine annealing phase. The warmup phase gradually increases the learning rate to help the model converge more effectively, while the cosine annealing phase gradually decreases the learning rate to fine-tune the model’s parameters[3].

Training and Validation:

The training process involves iterating over the training dataset in batches, computing the loss, performing backpropagation, and updating the model’s weights. The model is evaluated on the validation set after each epoch to monitor its performance and adjust the learning rate accordingly. The best-performing model based on validation accuracy is saved for future use. Each epoch includes processing batches of data, calculating loss, backpropagation, and weight updates. After each epoch, the model’s performance on the validation set is assessed, and the learning rate is adjusted. The model with the highest validation accuracy across epochs is considered the best[4].

Performance Evaluation:

The model’s performance is evaluated based on accuracy and loss metrics. The training and validation accuracy are tracked across epochs to assess the model’s learning progress and generalization capability. The best model weights are saved when the validation accuracy is maximized.

Integration with Classroom Knowledge

The methodology integrates several key concepts from machine learning and deep learning:

- **Data Augmentation:** A technique to artificially increase the size and diversity of the training dataset, reducing overfitting and improving model robustness.
- **Transfer Learning:** Utilizing a pre-trained model and fine-tuning it for a specific task, which is efficient and effective when labeled data is limited.
- **Optimizers:** Using SGD with momentum and weight decay to optimize the model's parameters effectively.
- **Learning Rate Scheduling:** Adjusting the learning rate dynamically to balance exploration and exploitation during training.
- **Model Evaluation:** Tracking accuracy and loss metrics to evaluate model performance and select the best model.

Critical Analysis

The chosen methodology demonstrates a comprehensive approach to image classification tasks, leveraging advanced techniques to enhance model performance. However, there are areas for potential improvement:

- **Hyperparameter Tuning:** The current settings for learning rate, momentum, and weight decay could be further optimized using techniques like grid search or Bayesian optimization.
- **Advanced Architectures:** Exploring newer architectures like Vision Transformers or EfficientNets could potentially yield better performance.
- **Regularization Techniques:** Incorporating more advanced regularization methods, such as batch normalization or weight noise, might further improve model generalization.

Overall, the methodology provides a solid foundation for image classification tasks, with room for further refinement and experimentation.

4 Experiments

Table 1 shows the training and validation accuracy across epochs. The model showed a steady increase in accuracy over the epochs, indicating effective learning.

Epoch	Training Accuracy	Validation Accuracy
1	0.7961	0.7941
2	0.8010	0.7961
$\vdots$	$\vdots$	$\vdots$
10	0.8010	0.7961

Table 1: Training and Validation Accuracy

References

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